

Stress tensor in NED Frame

$\{e_i\}$ geographic reference frame (NED)

$\{e'_i\}$ principal frame of stress tensor

Geophysicists can estimate

vertical stress: σ_v

min. horizontal stress: σ_h

max. horizontal stress: σ_H

There are the principal stresses: $\sigma_1 > \sigma_2 > \sigma_3$

Step 1: Order principal stresses

⇒ determines the type of fault

1) Normal Fault: $\sigma_1 = \sigma_v$ $\sigma_3 = \sigma_h$

2) Thrust Fault: $\sigma_1 = \sigma_H$ $\sigma_3 = \sigma_v$

3) Strike Slip: $\sigma_1 = \sigma_H$ $\sigma_3 = \sigma_h$

⇒ determines order and orientation
of principal directions $\{e_i\}$

Step 2: Principal directions in NED

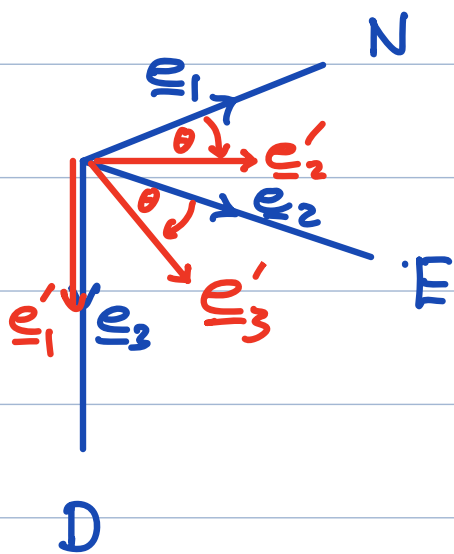
To find change in basis tensors

$$\underline{\underline{A}} = [\underline{\underline{e}}'_1 \ \underline{\underline{e}}'_2 \ \underline{\underline{e}}'_3]'$$

$[\underline{\underline{e}}'_i]$ = representation of $\underline{\underline{e}}'_i$ in NED frame

Note: Do not try to find $\underline{\underline{A}}$ directly
by guessing a rotation!

Example: Normal fault



$\sigma_1 = \sigma_v \longrightarrow \underline{\underline{e}}'_1$ is vertical

$\sigma_2 = \sigma_h$ (given azimuth θ)

$\sigma_3 = \sigma_h$

This may look like a
simple rotation by θ

around the vertical ($\underline{\underline{e}}_3$)

This would rotate $\underline{\underline{e}}_1$ into $\underline{\underline{e}}'_2$ instead of $\underline{\underline{e}}'_1$
 $\Rightarrow$ depends on ordering of σ 's (fault type)

Instead find $[\underline{e}_i']$ one by one.

Start with vertical stress in example

$$[\underline{e}_1'] \equiv \underline{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Continue with horizontal stress were you know the azimuth (σ_H in example) using a rotation

$$[\underline{e}_2'] = \underbrace{\underline{Q}(\underline{e}_3, \theta)}_{\substack{\uparrow \\ \text{rotates by } \theta \text{ around vertical (like } \underline{Q}_s)}} \underbrace{\underline{e}_1}_{\substack{\uparrow \\ \text{points north}}}$$

Find third $[\underline{e}_i']$ using 2nd frame identity

$$\underline{e}_i \times \underline{e}_j = \epsilon_{ijk} \underline{e}_k$$

$\Rightarrow$ this ensures right handed system (even perm.)

$$123 \rightarrow 231 \rightarrow 312$$

$$\text{Example: } [\underline{e}_1'] \times [\underline{e}_2'] = \cancel{\epsilon_{123}}^1 [\underline{e}_3']$$

Step 3: Find stress in NED frame

$$\underline{\underline{A}} = \begin{bmatrix} | & | & | \\ [\underline{e}_1'] & [\underline{e}_2'] & [\underline{e}_3'] \\ | & | & | \end{bmatrix}$$

Use change of basis for a tensor:

$$1) \quad [\underline{\underline{\sigma}}] = [\underline{\underline{A}}] [\underline{\underline{\sigma}}'] [\underline{\underline{A}}]^T$$

$$2) \quad [\underline{\underline{\sigma}}'] = [\underline{\underline{A}}]^T [\underline{\underline{\sigma}}] [\underline{\underline{A}}]$$

Given $[\underline{\underline{\sigma}}'] \rightarrow [\underline{\underline{\sigma}}]$ use 1